3 (a) (i) mean kinetic energy M1
of the atoms / molecules / particles A1
(ii) at absolute zero, atoms have no kinetic energy .... B1 [3]

(b) (i) $pV = nRT$ C1
$n = (1.2 \times 10^5 \times 2.0 \times 10^{-2}) / (8.31 \times 310)$ C1
$= 0.93$ mol A1
(ii) total amount = $(1.20 + 0.93)$ C1
$(1.20 + 0.93) = (4.0 \times 10^{-2} \times p) / (8.31 \times 310)$ C1
$p = 1.37 \times 10^5$ Pa .... A1 [6]